

CVIS Hands-on Training on Development of Machine Learning and Deep Learning Models Summer 2025

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Advanced Vision Tasks

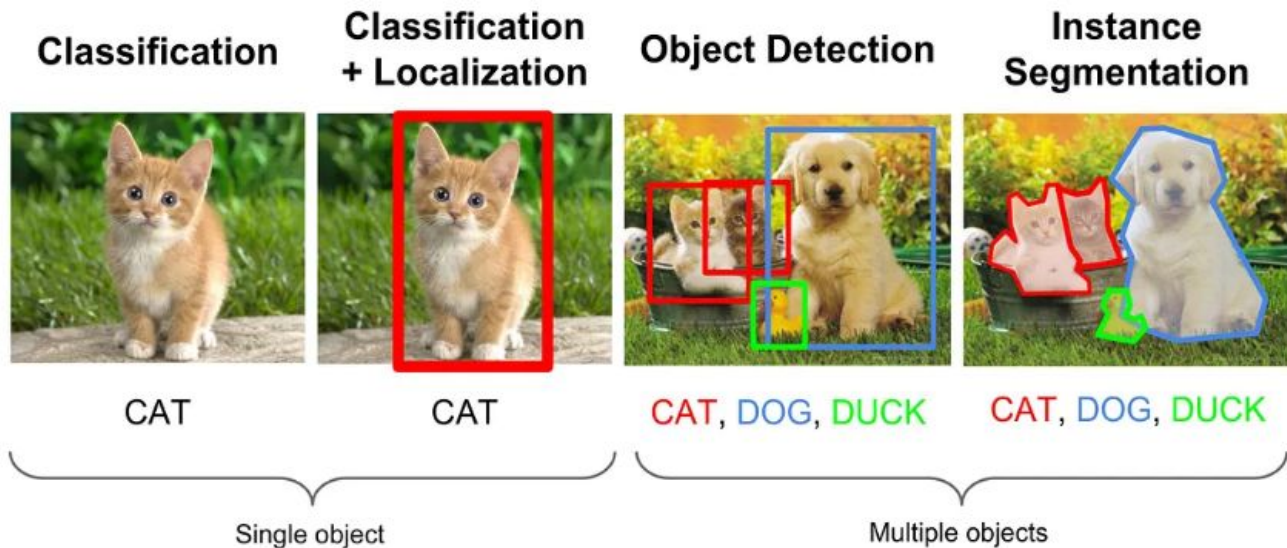
Object detection, Segmentation, Image annotation tools

01 Object Detection

Goal: Detect and Localize

Output: Bounding Boxes + Class Labels

<https://medium.com/zylapp/review-of-deep-learning-algorithms-for-object-detection-c1f3d437b852>
<https://www.v7labs.com/blog/object-detection-guide>



Comparison between image classification, object detection and instance segmentation.

Object Detection (OD) Category

Category	Examples	Pros	Cons
Two-stage detectors	Faster R-CNN, Cascade R-CNN	High accuracy, good for small objects	Slower, more complex to train
One-stage detectors	YOLO series, SSD, RetinaNet	Fast, real-time inference	Lower accuracy on small or overlapping objects
Transformer-based	DETR, DINO, RT-DETR	End-to-end training, global context awareness	Slow convergence, high memory usage
Diffusion-based	DiffusionDet, Pix2Seq	Uncertainty modeling, unified vision task	Still experimental, slower inference
Vision-language models	Grounding DINO, OWL-ViT, GLIP	Detect unseen categories, few-shot zero-shot support	Require large-scale training, less precise boxes
Lightweight models	YOLOv5n, MobileDet, NanoDet	Low-latency, energy efficient	Lower accuracy, poor generalization

Object Detection (OD) Category

Category	Detection Style	Key Characteristics	Best For
Two-Stage Detectors	Proposal → Refine	Generate region proposals first, then classify them	High-accuracy apps (e.g., medical, security)
One-Stage Detectors	Direct Prediction	Predict class + box in one pass over the image	Real-time use (e.g., self-driving, drones)
Transformer-Based	Attention-based	Replace convolutions with self-attention (DETR-style)	Scalable, open-set detection tasks
Diffusion-Based	Generative	Model detection as a denoising or generative task	Cutting-edge research, image synthesis
Vision-Language Models	Multimodal reasoning	Use text prompts to detect any object (open-vocab)	Generalist AI, VQA, robotics
Lightweight Models	Optimized CNNs	Architectures tailored for mobile or edge devices	Mobile apps, embedded devices



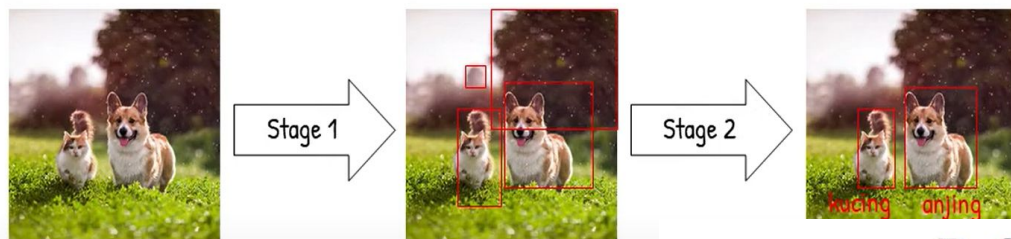
Glimpse of Each OD Category

1. Two-Stage Detectors: These models first generate region proposals, then classify them.

R-CNN, Fast R-CNN, Faster R-CNN, Cascade R-CNN, Libra R-CNN, Sparse R-CNN, DETR-RCNN (hybrid between RCNN and Transformers)

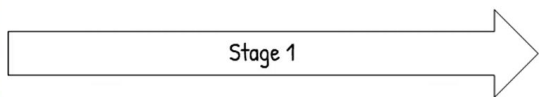
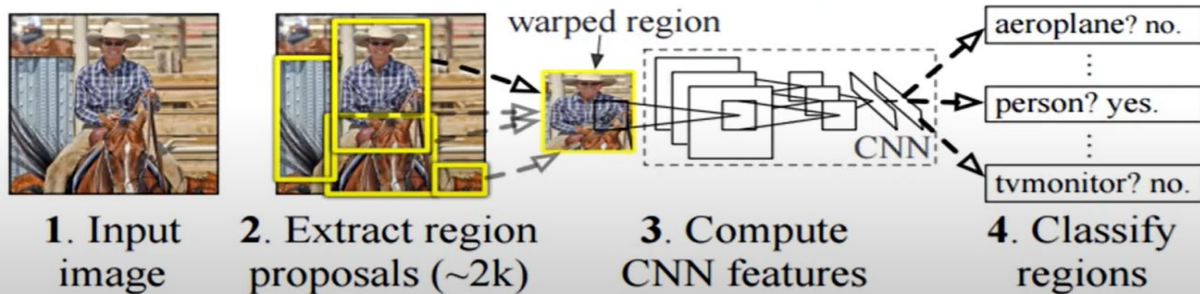
2. One-Stage Detectors: Single-step prediction, faster but initially less accurate (now competitive).

YOLO series, Single Shot MultiBox Detector (SSD), RetinaNet, EfficientDet, CenterNet, Fully Convolutional One-Stage (FCOS), YOLO-NAS, RT-DETR

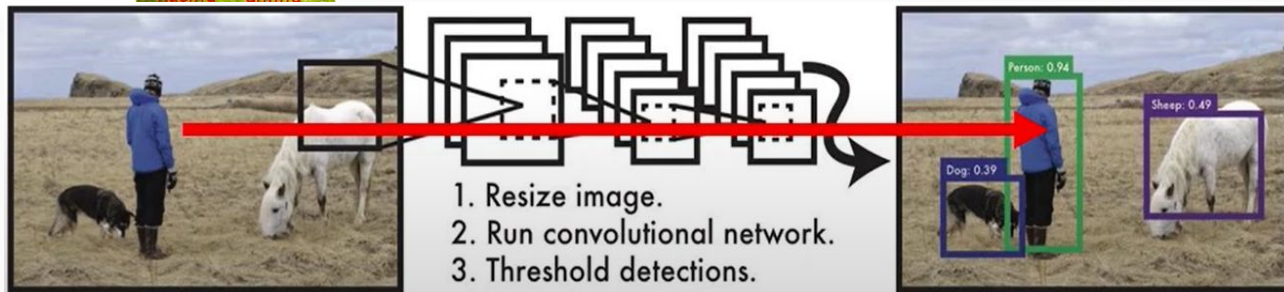


Two-Stage Detectors

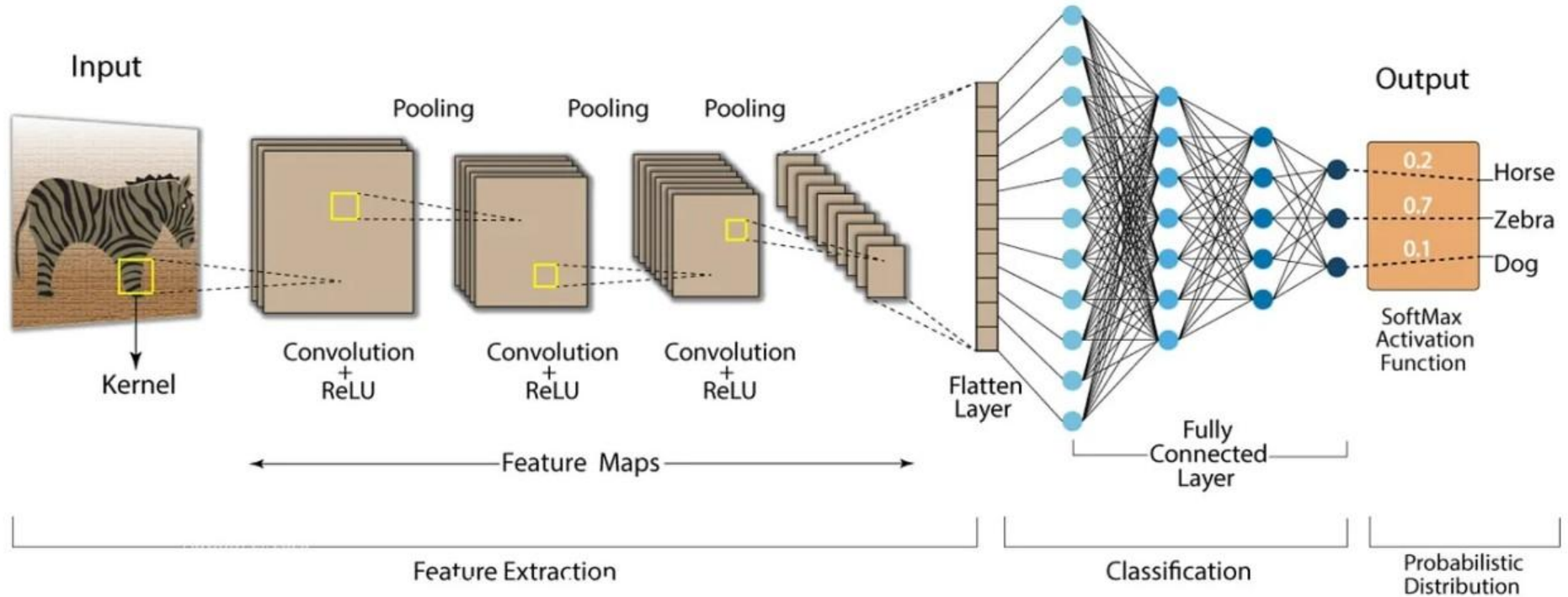
R-CNN: *Regions with CNN features*



One-Stage Detectors



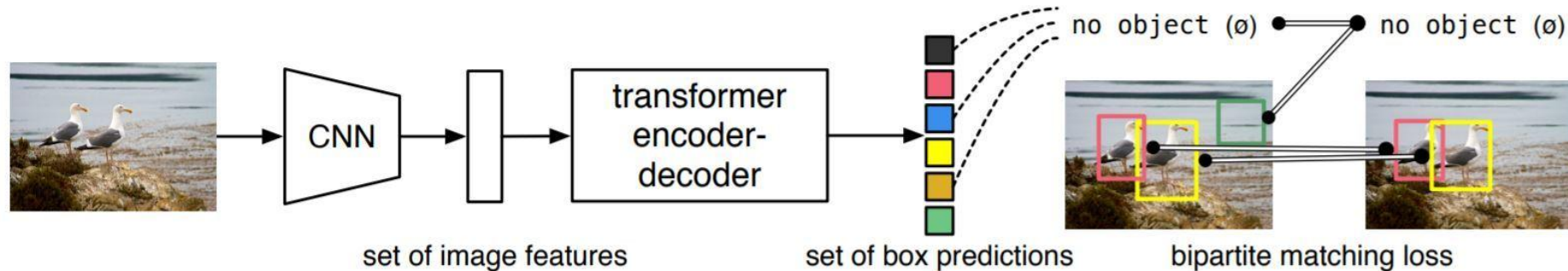
Convolution Neural Network (CNN)

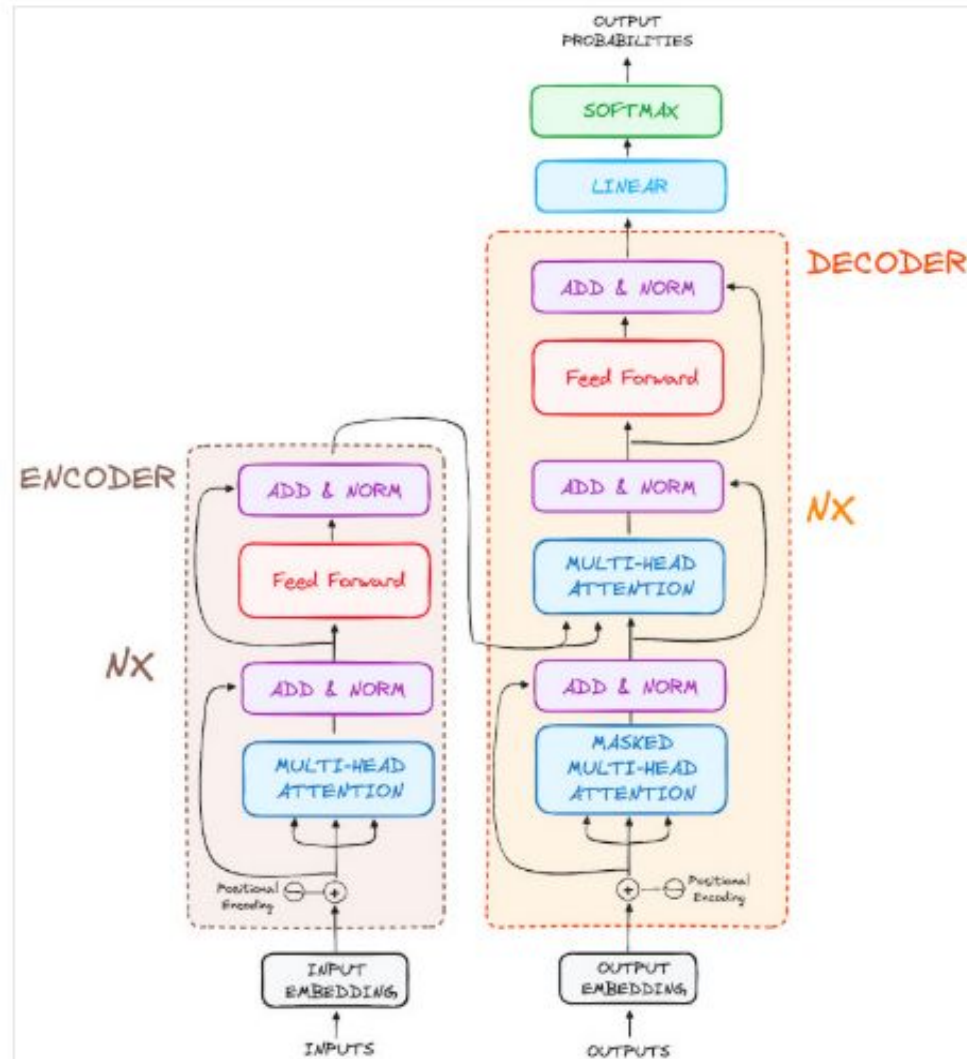


Glimpse of Each Category

3. Transformer-based Detectors: self-attention for end-to-end detection.

DETR, Deformable DETR, Conditional DETR, DN-DETR, DINO, Grounding DINO (GDINO), OWL-ViT / OWLv2, GLIP , Focal DETR





Transformer Architecture

<https://www.datacamp.com/tutorial/how-transformers-work>
Attention is All You Need



Glimpse of Each Category

4. Diffusion-Based & Generative Object Detectors: leverages generative models

DiffusionDet, DDOD, Pix2Seq, DINOv2 + Diffusion Adapter

5. Vision-Language & Foundation Models for Detection: These models work on unseen categories using language prompts.

OWL-ViT / OWLv2, GLIP, GDINO, SAM + DETIC, UniDet, Palo, Segment Everything Everywhere with Mask (SEEM)

6. Lightweight Models: Optimized for mobile or embedded devices

YOLO (nano or small), YOLO_Fastest, MobileDet, NanoDet, PP-YOLO Tiny



Object Detection Example Code

<https://github.com/Anika-Tahsin-S/Ship-Object-Detection-YoloV10/blob/main/ships-object-detection-using-yolov10.ipynb>

[Ultralytics YOLO11](#)

02

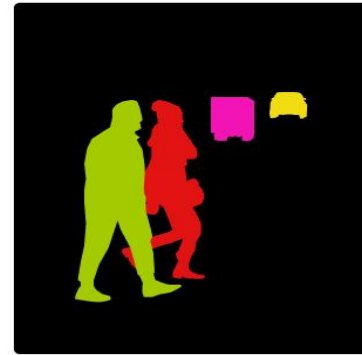
Segmentation

Goal: Assign a class to every pixel

Types of Image Segmentation



**SEMANTIC IMAGE
SEGMENTATION**



**INSTANCE
SEGMENTATION**



**PANOPTIC
SEGMENTATION**

<https://mindy-support.com/news-post/what-is-image-segmentation-the-basics-and-key-techniques/>



Read on Segmentation

[Using Multi-Scale Attention for Semantic Segmentation | NVIDIA Technical Blog](#)
[Beginner's Guide to Semantic Segmentation \[2024\]](#)
[What Is Instance Segmentation? \[2024 Guide & Tutorial\]](#)
[Panoptic Segmentation: Definition, Datasets & Tutorial \[2024\]](#)

Types of Segmentation

Type	Distinguishes Instances?	Labels Pixels?	Common Use-case	Pros	Cons
Semantic Segmentation	No	Yes	Scene parsing, road segmentation	Simpler	Cannot distinguish between individual object instances
Instance Segmentation	Yes	Yes	Object detection with pixel-level masks	Great for object counting, object tracking, and detailed detection	Computationally more expensive
Panoptic Segmentation	Yes (for objects), Yes (for background)	Yes	Unified scene understanding	Best for comprehensive scene understanding	More complex training and inference



Segmentation

1. Semantic Segmentation: Assigns a class label to every pixel in the image. All instances of a class are merged together

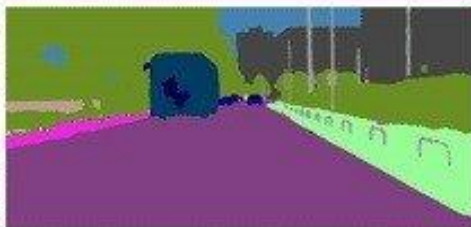
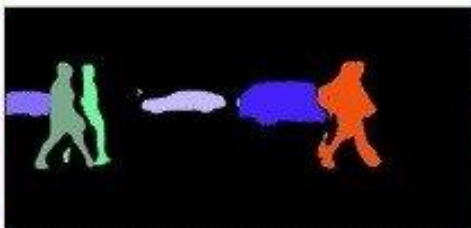
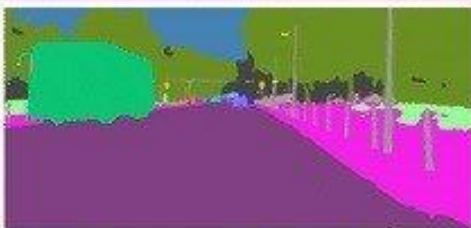
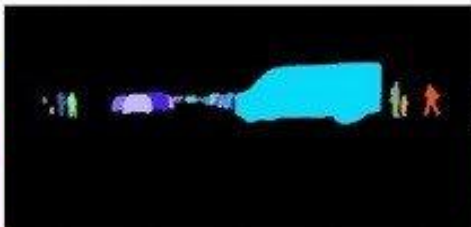
FCN, U-Net, DeepLab, SegNet, PSPNet

2. Instance Segmentation: Labels each pixel with both class and object instance. Differentiates each individual object, even if of the same class

Mask R-CNN, YOLACT, SOLO, BlendMask, Conditional Instance Segmentation (CondInst)

3. Panoptic Segmentation: 1 + 2

Panoptic FPN, UPSNet, Panoptic DeepLab, Mask2Former



Input

**Semantic Segmentation
Before Fusion**

**Instance Segmentation
Before Fusion**

Panoptic Segmentation



Advanced/Specialized Segmentations

Interactive / Weakly-Supervised Segmentation, 3D Segmentation, Video Segmentation, Real-time Segmentation, Weakly/Unsupervised Segmentation, Zero-shot Segmentation, Open-world Segmentation, Medical Segmentation, Multimodal Segmentation



Segmentation Example Code

Semantic Segmentation is Easy with Pytorch 😎



03 Image Annotation Tools

What Are They?

- Tools used to manually or semi-automatically label images with regions of interest (e.g., bounding boxes, masks, polygons).
- Essential for creating training data for object detection, segmentation, and classification tasks in computer vision.

Why Do We Use Them?

- Supervised Learning Needs Labels: Models require annotated data to learn visual patterns.
- Accuracy of Annotations = Model Performance: High-quality labels lead to better model predictions.
- Supports Multiple Tasks: Object detection → bounding boxes, Segmentation → masks/polygons, Keypoint detection → landmarks, Tracking → frame-wise object association

Tools Based on Tasks

Task	Recommended Tools
Object Detection (bbox)	Labellmg, CVAT, MakeSense.ai
Semantic Segmentation	CVAT, LabelMe, VIA, SuperAnnotate
Instance Segmentation	CVAT, Label Studio, SegUI (SAM)
Video Object Tracking	CVAT, VoTT, Scalabel
Medical Image Annotation	ITK-SNAP, 3D Slicer
Fast Prototyping	MakeSense.ai, VIA

Roboflow

Roboflow

<https://app.roboflow.com/ultimate-bun3e/semantic-segmentation-lrkwu/annotate/job/0QGDSA-VKlq5DEuCzha3u>

Label Studio

Label Studio

Please don't ask a question, Gotta go home!!

